

EXPERIMENT – 5

DESIGN AND SIMULATION OF UJT RELAXATION OSCILLATOR USING LTSPICE

OBJECTIVES

1. Analysis of UJT Relaxation Oscillator.
2. Determine Bandwidth
3. Determine Duty Cycle

EXPERIMENT – 5.1

FREQUENCY ANALYSIS OF UJT RELAXATION OSCILLATOR USING LTSPICE

After completion of this experiment, students will be able to:

1. Understand the construction and operating principle of a Unijunction Transistor (UJT).
2. Explain the working of a UJT Relaxation Oscillator.
3. Verify the generation of oscillations using a UJT-based oscillator circuit.
4. Observe the charging and discharging characteristics of the timing capacitor.
5. Measure the oscillation frequency and compare it with the theoretical value.
6. Analyze the effect of resistance and capacitance values on the oscillation frequency.
7. Identify and interpret the sawtooth waveform and trigger pulse waveform generated by the circuit.
8. Determine the role of the intrinsic stand-off ratio (η) in oscillator operation.
9. Plot and analyze the output waveforms using a CRO/oscilloscope or simulation tool.
10. Apply UJT oscillators in timing, triggering, pulse generation, and sweep generation applications.

Aim

To verify the operation of a UJT Relaxation Oscillator.

THEORY

The UJT relaxation oscillator generates non-sinusoidal oscillations.

Operation:

- Capacitor charges exponentially.
- UJT triggers at peak voltage.
- Capacitor discharges rapidly.
- Cycle repeats continuously.

DESIGN FORMULA:

Frequency

$$f = \frac{1}{RC \ln \left(\frac{1}{1-\eta} \right)}$$

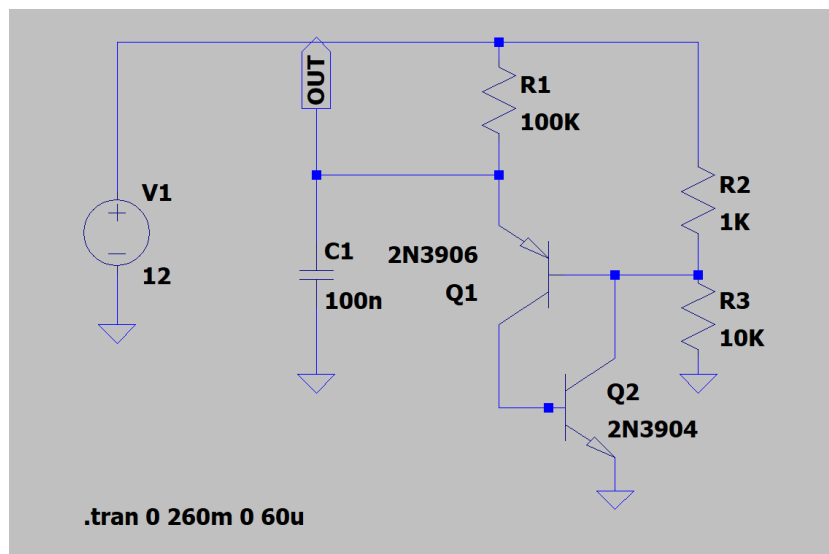
where

η = Intrinsic Stand-off Ratio

PROCEDURE

1. Draw UJT oscillator in LTSpice.
2. Configure RC network.
3. Run transient analysis.
4. Observe sawtooth waveform.
5. Measure oscillation frequency.

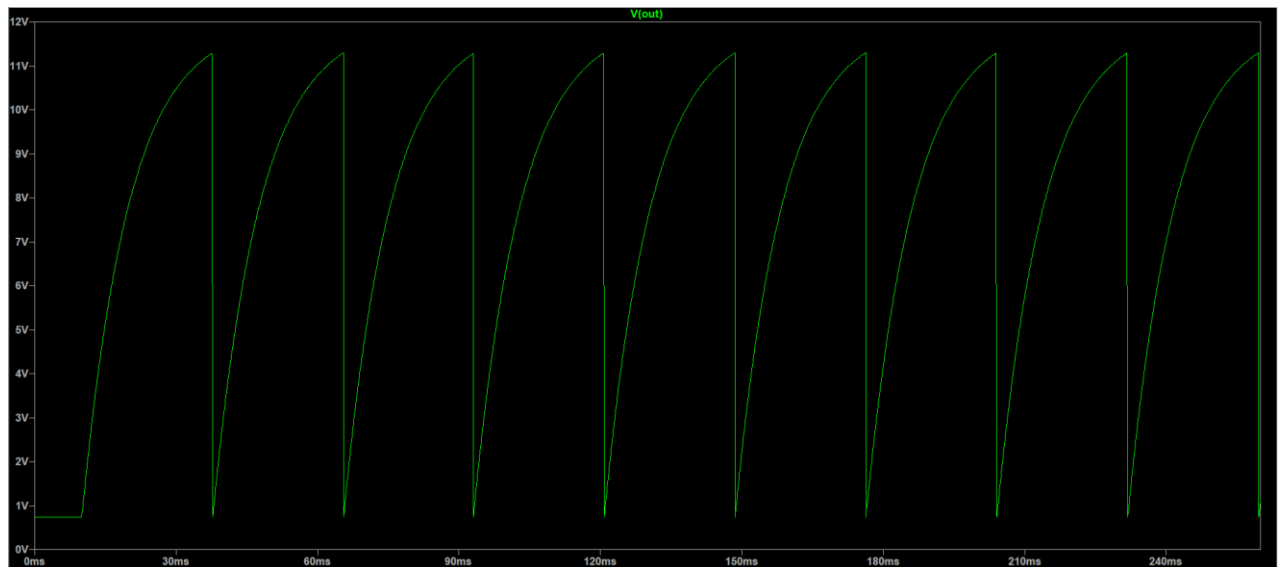
Circuit diagram:



Procedure:

1. Connections are made as shown in the circuit diagram.
2. The DC power supply is switched ON.
3. The output waveform is displayed on the CRO.
4. The peak-to-peak Amplitude and time period of the sine wave is noted.
5. The graph of output waveform is drawn.

Model Graph:



OBSERVATION TABLE

Waveform	Observation
Amplitude	11.3V
Time period	37.76ms
Frequency	26.48 Hz

Result:

UJT Relaxation Oscillator using LTSpice was completed and output obtained.

EXPERIMENT – 5.2

BANDWIDTH ANALYSIS OF UJT RELAXATION OSCILLATOR USING LTSPICE

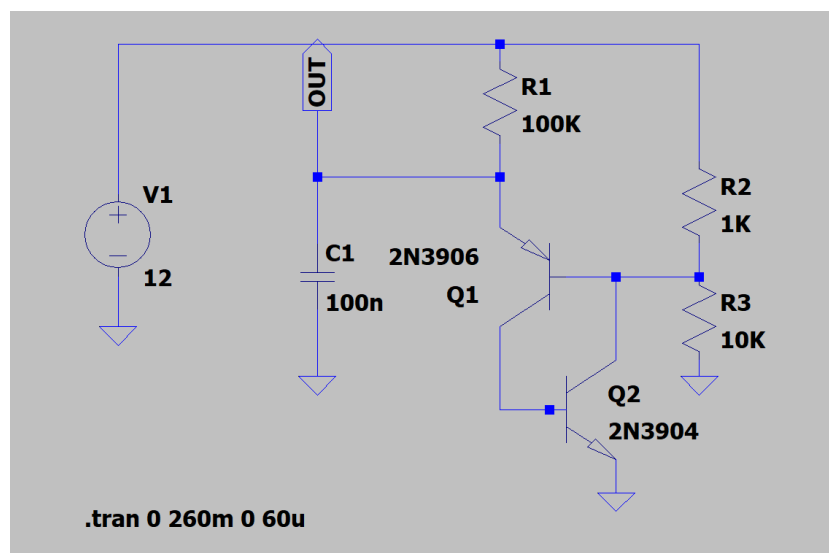
After completion of this experiment, students will be able to:

1. Understand the operating principle of a UJT Relaxation Oscillator and its frequency determining mechanism.
2. Simulate a UJT Relaxation Oscillator using LTSpice software.
3. Analyze the charging and discharging behavior of the timing capacitor in the oscillator circuit.
4. Measure the oscillation frequency from the simulated output waveform.
5. Compare theoretical and simulated frequency values of the oscillator.
6. Investigate the effect of varying resistance (R)(R)(R) and capacitance (C)(C)(C) values on the oscillation frequency.
7. Observe and interpret the generated sawtooth and pulse waveforms using LTSpice.
8. Plot and analyze the relationship between frequency and RC time constant.
9. Apply LTSpice simulation tools for frequency analysis and performance evaluation of relaxation oscillators.

Aim

To verify frequency variation with RC values.

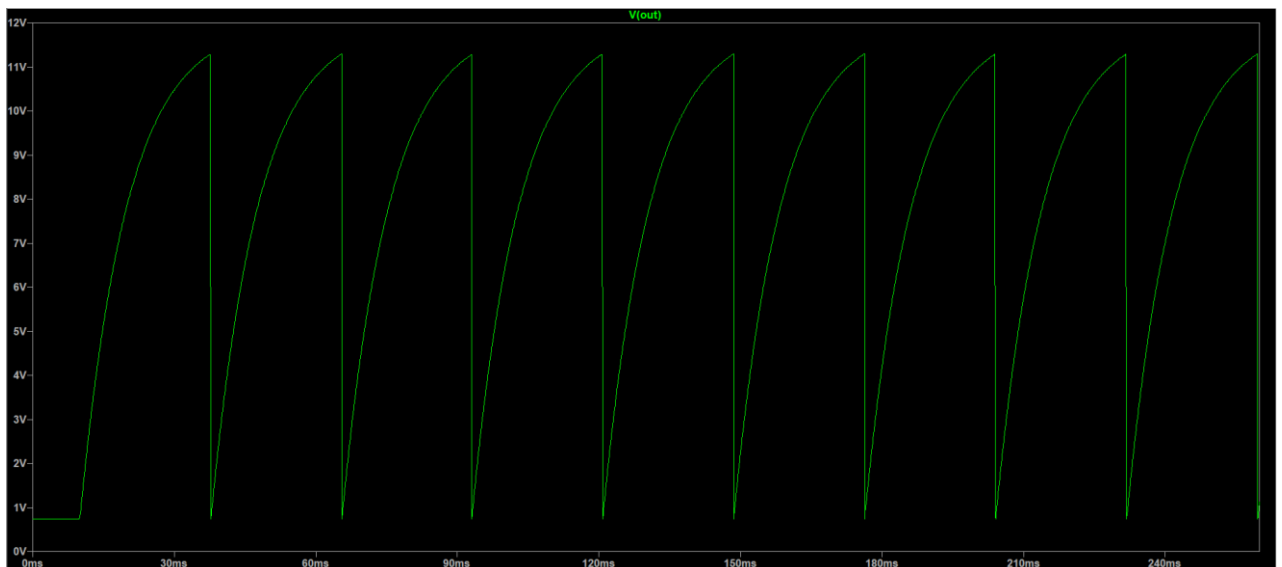
Circuit diagram:



Procedure:

1. Connections are made as shown in the circuit diagram.
2. The DC power supply is switched ON.
3. The output waveform is displayed on the CRO.
4. The peak-to-peak Amplitude and time period of the sine wave is noted.
5. The graph of output waveform is drawn.

Model Graph:



OBSERVATION TABLE

Waveform	Observation
Amplitude	11.3V
Time period	37.76ms
Frequency	26.48Hz
Bandwidth	36.04Hz

Result:

Thus the bandwidth of UJT Relaxation Oscillator using LTSpice was completed and output obtained.

EXPERIMENT – 5.3

WAVEFORM ANALYSIS OF UJT RELAXATION OSCILLATOR WITH DUTY CYCLE USING LTSPICE

After completion of this experiment, students will be able to:

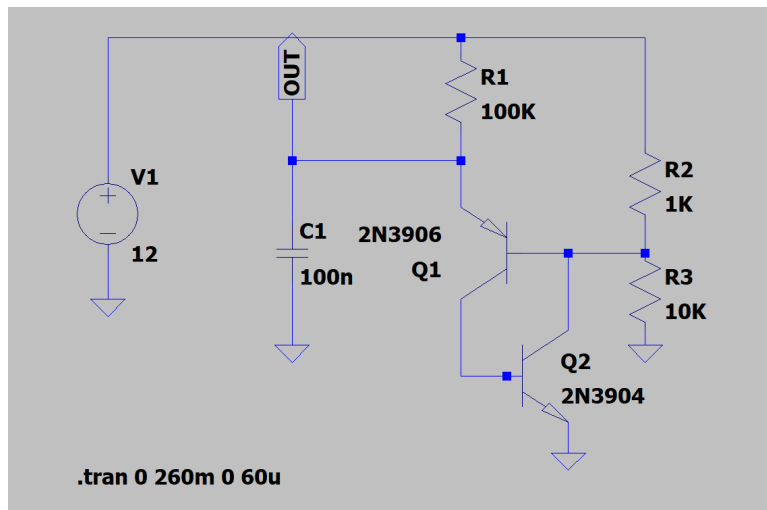
1. Understand the operating principle of a UJT Relaxation Oscillator and its frequencydetermining mechanism.
2. Simulate a UJT Relaxation Oscillator using LTSpice software.
3. Analyze the charging and discharging behavior of the timing capacitor in the oscillator circuit.
4. Measure the oscillation frequency from the simulated output waveform.
5. Compare theoretical and simulated frequency values of the oscillator.
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7. Observe and interpret the generated sawtooth and pulse waveforms using LTSpice.
8. Plot and analyze the relationship between frequency and RC time constant.
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Duty Cycle

- Ratio of pulse width to total period:

$$\text{Duty Cycle} = \frac{\text{Pulse Width}}{\text{Period}} \times 100\%$$

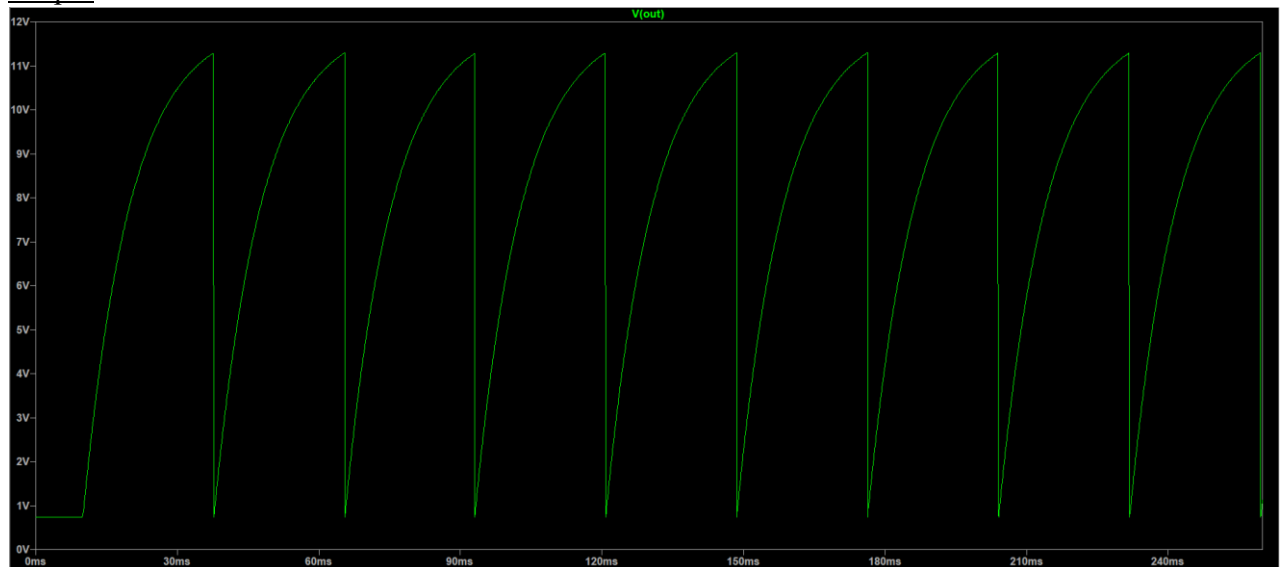
Circuit diagram:



Procedure:

1. Connections are made as shown in the circuit diagram.
2. The DC power supply is switched ON.
3. The output waveform is displayed on the CRO.
4. The peak-to-peak Amplitude and time period of the sine wave is noted.
5. The graph of output waveform is drawn.

Graph:



OBSERVATION TABLE

Waveform	THEORITICAL	PRACTICAL
Pulse width	28	27.741 ms
period	39	37.769 ms
Duty Cycle	73	73.44

Result:

UJT Relaxation Oscillator using LTSpice was completed and output obtained.